

DUET: Endogenous Failure Detection and Recovery via Dual-Action Manifold Alignment

Anonymous Authors

Abstract—Vision-language-action (VLA) policies map visual observations and language instructions to robot actions, but closed-loop execution can still end in failure after the robot drifts away from behavior that was successful in training. We focus on recoverable execution failures: cases in which the current rollout has become unreliable, yet re-entering the demonstrated behavior remains feasible. Such failures often begin as an execution deviation, where the observation, robot-object configuration, or action consequence leaves the support of successful demonstrations. We use this deviation as a task-specific form of execution out-of-distribution (OOD), rather than treating OOD and failure as equivalent. Existing solutions often add external failure detectors and collect human takeover or recovery data, making detection and control costly and loosely coupled. We introduce DUET (Dual-manifold Unified Execution and Transfer), a dual-action-head module that predicts relative actions for smooth nominal control and absolute waypoints for trajectory re-entry. Disagreement between the two action representations signals rising recoverable-failure risk, while the absolute waypoint-induced correction initiates a retry. Asymmetric training makes the relative branch sensitive to deviations and the absolute branch robust to visual disturbances. We evaluate DUET in LIBERO simulation and real-world manipulation under recoverable visual, viewpoint, scene, and action disturbances. DUET preserves nominal success (TODO% vs. TODO% for the relative-only baseline) while achieving TODO% retry success, outperforming the strongest baseline by TODO percentage points.

I. INTRODUCTION

Vision-language-action (VLA) policies adapt pretrained vision-language models to map visual observations and language instructions directly to robot actions [1], [2], [3]. They are attractive because one policy can connect semantic instructions to continuous manipulation. Yet a policy trained with limited robot data can still drift during closed-loop execution and eventually fail. An execution failure is a task-level outcome: the current rollout can no longer reliably complete the instruction. Before that outcome becomes irreversible, the robot may already have departed from the observations, robot-object configurations, or action consequences represented by successful demonstrations. We refer to this intermediate condition as an execution deviation, or execution OOD. OOD and failure are related but not equivalent: an unseen state may still be handled successfully, while a failure can also arise from compounding errors within seemingly familiar observations. Our focus is the practical regime in which an execution deviation raises failure risk but the task can still be completed by returning toward demonstrated behavior.

This paper is an ICRA 2026 draft. Author names and affiliations are omitted for anonymous review.

Mainstream approaches commonly introduce an external monitor, critic, verifier, or predictive model for failure detection [4]. When recovery is also required, they typically collect human takeover data in which an operator intervenes and demonstrates a corrective action [5], [6], [7]. This creates substantial data and deployment cost, especially when disturbances affect visual observations, object configurations, robot poses, or action execution. More importantly, a detector only indicates that execution has become abnormal; it does not specify how the controller should return to a behavior that can still complete the task. Detection and recovery are often trained as separate capabilities, so their decisions need not be consistent. We therefore ask whether a VLA can use its own action predictions to detect a recoverable execution failure and generate a retry direction from successful demonstrations alone, without failure labels or takeover demonstrations.

DUET is motivated by a complementary asymmetry between two action representations. Relative-action policies produce smooth and stable motion around a demonstrated trajectory, but once the robot or object state deviates, a local command provides no persistent target for returning to that behavior. Absolute-action policies are less suited to fine local control, yet their predicted waypoints can provide a global reference toward a reachable portion of the demonstrated trajectory. DUET therefore learns the same successful behavior in two action spaces: a relative head for nominal execution and an absolute head for trajectory re-entry. After transporting the absolute waypoint into the live robot frame, we compare it with the relative action. Their disagreement measures whether local control remains consistent with the global reference and serves as an endogenous signal of rising recoverable-failure risk. It is not intended as a universal OOD score; it is a task-specific inconsistency signal tied to successful behavior.

At inference time, DUET converts the absolute waypoint into a correction from the live pose and compares it with the relative action. A soft gate keeps relative control in the nominal regime and shifts toward the waypoint-induced correction when disagreement indicates a recoverable execution failure, allowing the robot to re-approach demonstrated behavior and retry. The gate returns to relative control after the two predictions realign. DUET is therefore a failure-aware retry mechanism rather than a general recovery planner: it does not claim to solve irreversible failures, unreachable dropped objects, hardware faults, or cases that require discovering a new task strategy.

Our contributions are threefold:

- We formulate failure-aware retry as detecting recover-

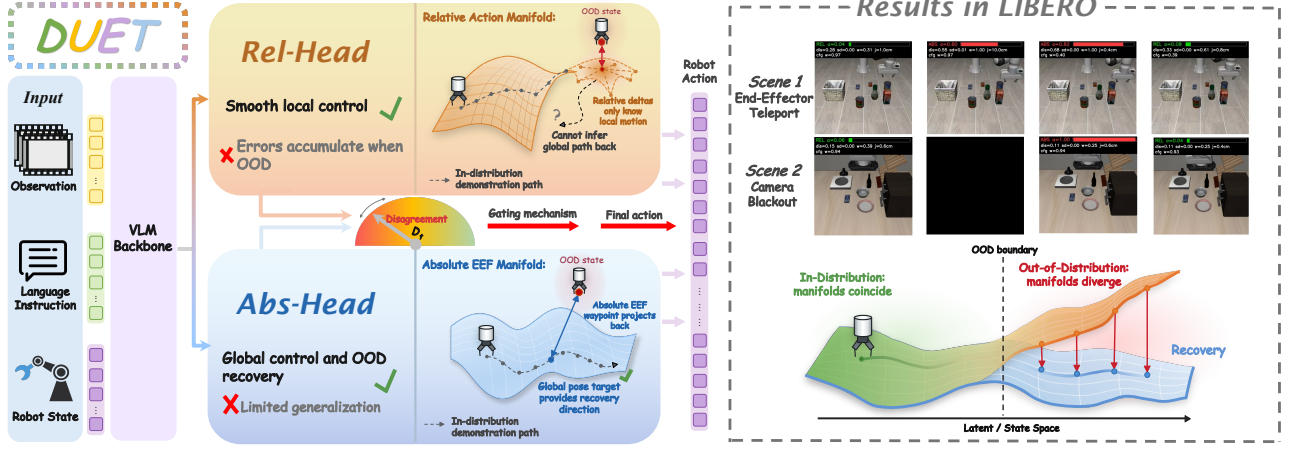


Fig. 1. Failure-aware retry with dual action spaces. Relative and absolute predictions agree during nominal execution but diverge when a recoverable deviation drives the rollout OOD, exposing an endogenous disagreement score D_t . Relative actions provide precise local motion, whereas absolute waypoints bring the robot back toward a demonstrated execution path for retry.

able execution deviations from successful behavior and re-entering its support, distinguishing this problem from irreversible failure recovery and task-level replanning.

- We propose DUET, a dual-action-head VLA that uses endogenous disagreement between relative and absolute action representations to detect rising recoverable-failure risk and activate an absolute waypoint-based retry while preserving smooth nominal control.
- We evaluate DUET in simulation and on a real robot across diverse visual, viewpoint, scene, and action disturbances, measuring both nominal robustness and retry success without additional human takeover data.

II. RELATED WORK

A. Failure Detection for Robot Policies

Runtime failure detection estimates whether a policy remains within the support of successful behavior. Diff-Dagger uses diffusion uncertainty to request expert intervention [8], while Sentinel monitors action inconsistency and task progress [9]. FAIL-Detect formulates failure detection as sequential OOD testing using only successful executions [4]; FIPER similarly combines representation novelty with action uncertainty without failure data [10].

VLA-specific methods use internal features or external foundation models. SAFE learns a detector from successful and failed rollouts [11], whereas AHA trains a VLM to detect and diagnose failures [12]. Such methods require additional supervision or a separate detector whose output is not a recovery action. DUET instead detects failures through disagreement between two action spaces learned within the policy, using OOD manifold inconsistency as both an alarm and a recovery signal.

B. Recovery from Execution Failures

Recovery methods commonly expand the training distribution with expert corrections. DAgger queries experts on policy-induced states [5]; intervention-based methods collect human takeovers [13], [6]; and IntervGen transforms

limited interventions into corrective trajectories [7]. These approaches reduce compounding errors but still require additional recovery actions.

Foundation-model systems instead use semantic supervisors. RACER learns from language-annotated recovery trajectories [14], while AHA and VLM controllers diagnose failures and guide downstream correction [12], [15]. Their detection and recovery components remain separately trained or prompted. DUET couples both capabilities without extra recovery data: absolute states from successful demonstrations provide recovery anchors, while cross-space disagreement determines when and how strongly to move toward them.

III. METHOD

A. Problem Formulation

At time t , the robot observes multi-view images o_t , a language instruction l , and its current proprioceptive state s_t . DUET is trained only on successful demonstrations and predicts the demonstrated behavior in two action representations over a horizon K :

$$\hat{\Delta}A_t = f_{\text{rel}}(o_t, l, s_t) \in \mathbb{R}^{K \times 7}, \quad (1)$$

$$\hat{W}_t = f_{\text{abs}}(o_t, l) \in \mathbb{R}^{K \times 7}. \quad (2)$$

Here, ΔA_t contains relative operational-space-control commands and W_t contains future absolute end-effector waypoints $[x, y, z, r_x, r_y, r_z, g]$. We decompose each command as $a_t = [a_{p,t}, a_{g,t}]$, where $a_{p,t} \in \mathbb{R}^6$ controls end-effector pose and $a_{g,t} \in \mathbb{R}$ controls the gripper. A successful trajectory naturally supplies both labels: its recorded delta actions supervise f_{rel} , while its future end-effector states supervise f_{abs} . The geometric analysis and fusion operate on a_p ; although the absolute head also predicts gripper state during training, the deployed a_g always comes from the relative head.

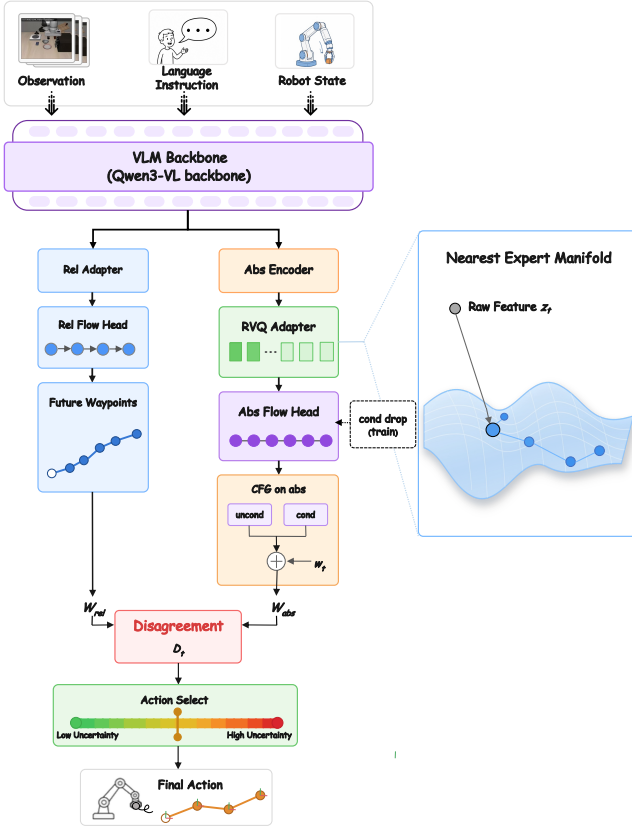


Fig. 2. DUET in four stages: a shared vision-language encoder, relative and absolute action heads, cross-space failure detection, and closed-loop recovery. The relative head provides state-aware local control; the explicitly state-free absolute head uses RVQ-anchored conditions to predict global waypoints. Their disagreement determines the transition between nominal execution and recovery.

B. Dual Action Manifolds

For an instruction l , let $\mathcal{M}_l \subset SE(3)$ denote the end-effector state manifold traced by successful demonstrations, and let $T\mathcal{M}_l$ denote its tangent bundle. Under nominal execution, the relative head predicts a local motion vector

$$\hat{\delta}_{a,p,t} = f_{\text{rel}}(o_t, l, s_t)_p \in T_{p_t}\mathcal{M}_l, \quad (3)$$

where p_t is the current end-effector pose. This tangent representation is locally smooth and expressive, but it specifies only how to move from the current chart. Once a failure displaces the robot from the demonstrated support, the extrapolated vector field contains no explicit destination to which the robot should return.

The absolute head instead predicts a future coordinate

$$\hat{w}_{p,t} = f_{\text{abs}}(o_t, l)_p \in \mathcal{M}_l, \quad (4)$$

which acts as a visual-language-conditioned global anchor. To compare the two representations, we introduce a state-dependent transport map

$$c_t = \Psi_{p_t}(\hat{w}_{p,t}) \in T_{p_t}SE(3), \quad (5)$$

which converts the absolute waypoint into a local correction at the live pose. Along an in-domain trajectory, c_t and $\hat{\delta}_{a,p,t}$

describe the same demonstrated motion in two coordinate systems and should therefore align. When a failure drives execution OOD, the relative head extrapolates a local field while the absolute head remains anchored to a demonstrated destination, exposing a cross-space inconsistency.

This geometry couples detection and recovery without asserting a formal OOD guarantee: the norm of the inconsistency provides a scalar failure signal, whereas the transported vector c_t already provides the corresponding recovery direction. DUET can therefore change both when and where the controller moves using the same pair of predictions.

C. Dual-Head Architecture

Figure 2 shows the shared VLM and two flow-matching action heads [16], [17]. The relative head attends to the full hidden sequence and explicitly receives s_t through a numerical state encoder. It is therefore state-aware and serves as the primary controller during nominal execution. The absolute head receives no explicit current proprioception. When state tokens are appended to the causal VLM prompt, they are placed last and excluded from absolute-branch pooling, while the relative head can still attend to them. This separation prevents the absolute predictor from collapsing into an integration of the current state and preserves its role as a global reference.

The absolute branch pools its state-free hidden sequence H^0 and applies residual vector quantization (RVQ) [18], [19]:

$$Z_q = \text{RVQ}(\text{Pool}(H^0)). \quad (6)$$

RVQ maps continuous conditions to a discrete atlas learned from successful demonstrations, anchoring absolute predictions to recurrent visual-language modes rather than an unconstrained feature vector. Its residual norm is retained only as auxiliary telemetry and is not part of the primary failure score. The absolute flow head additionally uses classifier-free guidance (CFG) [20], enabling its conditional prediction to be softened toward the demonstrated waypoint marginal during recovery.

D. Disagreement-Preserving Learning

The two heads are supervised by the paired labels available in every successful trajectory. Both use flow-matching objectives, while RVQ contributes a commitment loss. The complete default objective is

$$\mathcal{L} = \lambda_{\text{rel}}\mathcal{L}_{\text{rel}} + \lambda_{\text{abs}}\mathcal{L}_{\text{abs}} + \lambda_{\text{vq}}\mathcal{L}_{\text{vq}}, \quad (7)$$

where the absolute labels are normalized future states and the relative labels are the recorded delta commands. We deliberately impose no cross-head consistency loss or distillation: forcing agreement everywhere would suppress the inconsistency needed for failure detection.

DUET instead specializes the branches through asymmetric visual augmentation. The absolute head and RVQ adapter train on both clean and perturbed observations, encouraging

the global anchor to remain stable under degraded input. The relative head is optimized only on the clean subset $\mathcal{B}_{\text{clean}}$:

$$\mathcal{L}_{\text{rel}} = \frac{1}{|\mathcal{B}_{\text{clean}}|} \sum_{i \in \mathcal{B}_{\text{clean}}} \ell_{\text{FM}}(\hat{\Delta} a^{(i)}, \Delta a^{(i)}). \quad (8)$$

This gradient truncation preserves relative-head sensitivity while training the absolute branch as the robust reference, making cross-space disagreement more discriminative under visual OOD. For CFG training, the absolute head also replaces its quantized condition with learned null tokens on a subset of samples and learns both conditional and marginal waypoint flows.

E. Joint Failure Detection and Recovery

At deployment, the policy returns both unnormalized action chunks. Let the live pose be $p_t = (x_t, R_t)$ and the pose component of the current absolute waypoint be $\hat{w}_{p,t} = (\hat{x}_t, \hat{R}_t)$. DUET implements the transport map in Eq. (5) as

$$c_t = \text{clip} \left(\begin{bmatrix} (\hat{x}_t - x_t)/\sigma_p \\ \text{Log}(\hat{R}_t R_t^{-1})/\sigma_R \end{bmatrix}, -1, 1 \right), \quad (9)$$

where $\text{Log} : SO(3) \rightarrow \mathfrak{so}(3)$ gives the geodesic rotation correction and σ_p, σ_R convert pose error into action units. The cross-space disagreement is

$$D_t = \frac{\|c_t - \text{clip}(\hat{\delta} a_{p,t}, -1, 1)\|_2}{\sqrt{6}}. \quad (10)$$

This is the primary failure signal. We supplement it with a coarse state-distance term that measures normalized violation of the training-state bounds:

$$S_t = \text{mean} \left[\frac{\max(s_{\min} - p_t, 0)}{s_{\max} - s_{\min}} + \frac{\max(p_t - s_{\max}, 0)}{s_{\max} - s_{\min}} \right]. \quad (11)$$

The resulting failure score and recovery weight are

$$q_t = D_t + \lambda S_t, \quad (12)$$

$$\alpha_t = \sigma(k(q_t - \tau)). \quad (13)$$

DUET then interpolates between local execution and global recovery while retaining relative gripper control:

$$a_t = [a_{p,t}, a_{g,t}], \quad (14)$$

$$a_{p,t} = (1 - \alpha_t) \hat{\delta} a_{p,t} + \alpha_t c_t, \quad (15)$$

$$a_{g,t} = \hat{\delta} a_{g,t}, \quad (16)$$

where $[\hat{\delta} a_{p,t}, \hat{\delta} a_{g,t}]$ is the relative-head command. Thus relative control dominates along the demonstrated support, while a rising failure score shifts a_p toward the absolute recovery direction; control returns smoothly after the two representations realign. When requesting a new chunk, the previous fusion weight also softens the absolute-head CFG scale,

$$w_t = w_{\max} - (w_{\max} - w_{\min}) \alpha_{t-1}. \quad (17)$$

The disagreement D_t therefore determines *when* recovery is needed, while the vector c_t determines *how* to recover.

IV. EXPERIMENTS

We organize the experiments around three claims:

- (c1) **Anomaly detection:** Can DUET identify execution anomalies from successful demonstrations alone?
- (c2) **Nominal execution:** Does the dual-action mechanism preserve normal task execution?
- (c3) **Retry recovery:** Can DUET use the detected anomaly to re-enter successful behavior and complete the task?

We evaluate these claims with matched LIBERO experiments, controlled execution disturbances, detection-score analysis, component ablations, and a reserved real-robot protocol.

A. Evaluation Protocol

We evaluate DUET on the four LIBERO suites—Spatial, Object, Goal, and LIBERO-10—with 10 tasks and 50 initial states per task. DUET and DUET-Rel use the same suite-specific checkpoint, training data, initial states, and disturbance seeds; DUET-Rel fixes $\alpha_t = 0$ and uses only the relative branch. Unless otherwise stated, we report paired closed-loop task success. When a disturbance has a semantic precondition, we also report its injection coverage.

B. (c1) Anomaly Detection

Claim (c1) asks whether disagreement between relative and absolute actions can identify execution anomalies without failure examples. We compare DUET’s runtime score with detectors that likewise require no failure examples: logpZO, RND, and PCA-kmeans from the FAIL-Detect framework [4], and STAC, which scores temporal inconsistency between overlapping action chunks [9]. Table I reports discrimination and temporal localization under the same protocol.

DUET attains the highest TPR (0.900) and AUROC (0.845) among detectors trained without failure examples. This supports (c1): cross-action disagreement identifies when execution becomes inconsistent with successful behavior. Because the score drives a soft controller gate rather than a binary interrupt, sensitivity and ranking are the relevant detection properties.

C. (c2) Nominal Execution

Claim (c2) asks whether failure-aware control preserves ordinary execution when no disturbance is injected. Table II evaluates this claim. DUET attains 96.9% average success, matching the reported $\pi_{0.5}$ result and slightly exceeding the reported StarVLA-GR00T average. External rows provide benchmark context; the controlled comparison is DUET versus DUET-Rel. Adaptive fusion raises the average from 93.8% to 96.9%, with gains on all four suites and the largest lifts on Goal (+5.6) and LIBERO-10 (+5.8). This supports (c2): adaptive dual-action control does not compromise nominal task completion.

TABLE I

FAILURE-DETECTION COMPARISON. TPR, TNR, BALANCED ACCURACY (BAL.-ACC.), AUROC, AND LOC.@16 ARE HIGHER-IS-BETTER; DETECTION DELAY IS LOWER-IS-BETTER. BOLD VALUES INDICATE THE BEST RESULT IN EACH COLUMN.

Method	TPR $\uparrow$	TNR $\uparrow$	Bal.-Acc. $\uparrow$	AUROC $\uparrow$	Delay (steps) $\downarrow$	Delay (s) $\downarrow$	Loc.@16 $\uparrow$
DUET	0.900	0.265	0.582	0.845	38.3	1.91	0.523
logpZO [4]	0.889	0.243	0.566	0.817	22.7	1.14	0.569
RND [21], [4]	0.842	0.361	0.602	0.549	26.1	1.31	0.330
PCA-kmeans [4]	0.884	0.283	0.583	0.625	20.3	1.01	0.503
STAC [9]	0.839	0.311	0.575	0.494	25.8	1.29	0.260

TABLE II

NOMINAL TASK SUCCESS (%) ON LIBERO. EXTERNAL ROWS ARE REPORTED RESULTS AND ARE INCLUDED ONLY FOR BENCHMARK CONTEXT. STARVLA USES JOINT 30K TRAINING; DUET USES SUITE-SPECIFIC 10K CHECKPOINTS.

Method	Spatial	Object	Goal	LIBERO-10	Avg.
$\pi_{0.5}$ (reported)	98.8	98.2	98.0	92.4	96.9
StarVLA-GR00T (reported)	97.8	98.8	97.4	92.0	96.5
DUET	98.8	99.4	97.6	91.8	96.9
DUET-Rel	98.0	99.2	92.0	86.0	93.8

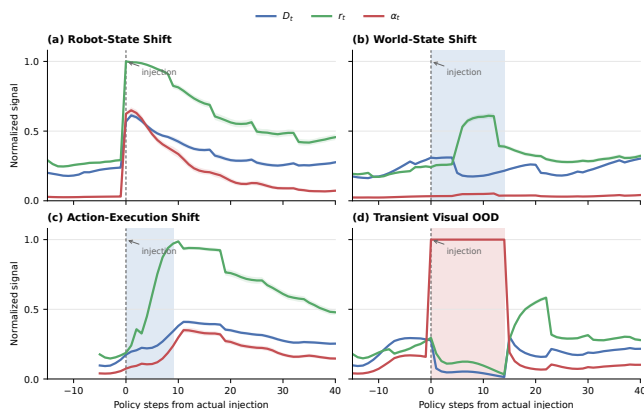


Fig. 3. Controller dynamics on the Object suite under (a) end-effector teleport, (b) vertical grasp drop, (c) action perturbation, and (d) camera blackout. Traces are aligned to the actual injection time and show means with pointwise bootstrap 95% confidence intervals over successfully injected episodes. Shading denotes the perturbation or forced-hold window.

D. (c3) Retry Recovery under Execution Disturbances

Claim (c3) asks whether DUET can convert an execution anomaly into a successful retry. We evaluate closed-loop task success after controlled disturbances that leave the task objective recoverable. Table III summarizes the results. In every aggregate row DUET matches or exceeds DUET-Rel. The largest gains occur precisely where the introduction predicts relative control to be insufficient: once locally plausible commands have driven the trajectory OOD, or once the arm has been displaced from the demonstrated pose.

1) *Action-Execution Failures*: We first corrupt the executed 6D command over a short window. The scene is unchanged, but execution error accumulates and the induced trajectory leaves the demonstrated support—the setting in

which a relative policy can keep issuing locally plausible commands without a return target. The evaluator provides no perturbation flag; any switch must come from the two action predictions.

This is the strongest closed-loop result. Over eight suite-severity groups (4000 paired episodes), DUET reaches 64.6% success versus 27.9% for DUET-Rel (+36.7 points; Table IV). Every cell improves, and the gap widens as relative control collapses: under severe perturbations DUET-Rel falls to 5.4–12.8%, while DUET retains 43.2–60.6%, including +52.8 points on Object. Figure 3(c) shows disagreement and the fusion weight rising as the arm drifts, then returning toward relative control after re-entry.

2) *Robot-Arm Displacement*: We next teleport the end effector while leaving the scene unchanged, corresponding to an unexpected arm displacement. Relative commands remain locally defined around the off-trajectory pose; the absolute head still predicts a demonstrated waypoint and supplies a retry direction.

Across 2000 paired episodes, DUET improves success from 40.0% to 61.7% (+21.7 points; Table IV). Object improves by +39.8 points, LIBERO-10 by +25.0, and Goal by +19.0. Among Object episodes with a successful injection, DUET completes 72.8% versus 33.2% for DUET-Rel. Figure 4 shows the intended switch-and-return: relative control before the displacement, absolute correction immediately afterward, and a return to the relative head after re-entry. The Spatial difference is smaller (+3.0 points, $p=0.21$), in part because about 12% of episodes finish before the scheduled teleport.

3) *Object and Grasp Disturbances*: A vertical grasp drop changes the robot-object configuration rather than the arm pose: at policy step 67, if a free-joint object is in gripper contact, the evaluator opens the gripper, keeps the object’s XY position, lowers it by 5 cm, and holds the gripper open for 15 steps. Episodes with no grasped object are marked `not_grasped` and left unperturbed.

Object is the suite where this event is well defined. DUET succeeds in 476/500 episodes (95.2%) versus 446/500 (89.2%) for DUET-Rel (+6.0 points; bootstrap 95% CI [3.4, 8.8]; McNemar $p=2.4 \times 10^{-5}$). Injection occurs in 498/500 DUET and 494/500 DUET-Rel episodes. The absolute gain is smaller than under action or teleport disturbances because DUET-Rel already remains near 89%; the direction is the same as in the introduction’s dropped-object

TABLE III

CLOSED-LOOP TASK SUCCESS UNDER MATCHED EXECUTION-FAILURE PROTOCOLS. EACH ROW USES ONLY THE STATED SCOPE.

Family	Protocol	Scope	Pairs	DUET-Rel	DUET	Δ (pp)
Action execution	Action perturbation	4 suites, medium+severe	4000	27.9	64.6	+36.7
Robot displacement	EEF teleport	4 suites, medium	2000	40.0	61.7	+21.7
Visual occlusion	Camera blackout	4 suites, step 15 medium	2000	83.4	92.2	+8.8
Visual occlusion	Camera blackout	4 suites, step 23 medium+severe	4000	73.5	90.8	+17.3
Object / grasp	Vertical grasp drop	Object, medium	500	89.2	95.2	+6.0
Contact / scene	External force	4 suites, medium+severe	4000	92.22	95.95	+3.73

TABLE IV

CLOSED-LOOP TASK SUCCESS (%) UNDER ACTION, ROBOT-POSE, AND OBJECT/GRASP DISTURBANCES. DUET AND DUET-REL USE THE SAME CHECKPOINT. $N=500$ PER CELL.

Setting	Suite	DUET-Rel	DUET	Δ	p
<i>Action-execution shift</i> (action perturbation, step 5, $N=500$)					
medium	Spatial	54.0	74.0	+20.0	1×10^{-14}
medium	Object	48.8	84.4	+35.6	1×10^{-34}
medium	Goal	47.2	74.2	+27.0	5×10^{-22}
medium	LIBERO-10	36.6	78.6	+42.0	1×10^{-48}
severe	Spatial	12.8	43.2	+30.4	6×10^{-31}
severe	Object	7.8	60.6	+52.8	1×10^{-70}
severe	Goal	10.6	52.6	+42.0	3×10^{-49}
severe	LIBERO-10	5.4	49.0	+43.6	2×10^{-53}
	Overall	27.9	64.6	+36.7	—
<i>Robot-arm displacement</i> (end-effector teleport, medium, $N=500$)					
	Spatial	54.6	57.6	+3.0	0.21
	Object	36.2	76.0	+39.8	1×10^{-40}
	Goal	33.4	52.4	+19.0	2×10^{-12}
	LIBERO-10	35.8	60.8	+25.0	1×10^{-17}
	Overall	40.0	61.7	+21.7	—
<i>Object / grasp disturbance</i> (vertical grasp drop, Object, $N=500$)					
	Object	89.2	95.2	+6.0	2.4×10^{-5}

example. On Goal, both methods receive the drop in only 282/500 paired episodes, so we do not treat Goal as a suite-level object-disturbance result. Among those both-injected episodes the success rates are 23.4% for DUET and 17.7% for DUET-Rel.

4) *Transient Visual Occlusion*: A complete camera blackout removes the current visual observation. During the blackout, DUET holds the last reliable absolute waypoint and resumes dual-action control when vision returns, using the global action stream as short-term execution memory.

On the matched step-15 medium protocol, success rises from 83.4% to 92.2% (+8.8 points; Table V). LIBERO-10 accounts for the largest suite-level gain (+22.4 points); Object is already at ceiling (97.4% vs. 97.0%, $p=0.84$). Under the longer step-23 medium+severe protocol the gap widens to +17.3 points (90.8% vs. 73.5%). A longer visual outage therefore increases, rather than removes, the value of the absolute waypoint. Figures 3(d) and 5 show that the cached waypoint is used during the blackout and that adaptive control resumes afterward. The Goal task “open the top drawer and put the bowl inside” remains a counterexample at step 15: a contact sequence that still needs updated visual feedback is outside the retry mechanism claimed in

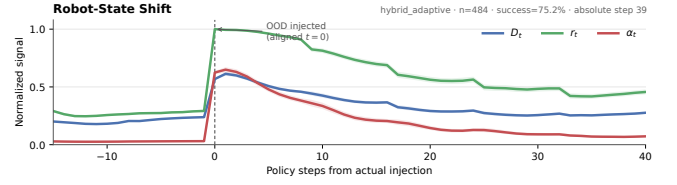


Fig. 4. End-effector teleport on the Object suite. Top: a successful retry, with relative control before the displacement, absolute waypoint correction immediately afterward, and a return to relative control after re-entry. Bottom: mean disagreement D_t and fusion weight α_t over 484 successfully injected episodes, aligned to the injection time; bands are pointwise bootstrap 95% confidence intervals.

TABLE V

TASK SUCCESS (%) UNDER CAMERA BLACKOUT. STEP-15 MEDIUM USES 2000 PAIRED EPISODES; STEP-23 AGGREGATES EIGHT SUITE-SEVERITY GROUPS.

Protocol	Suite	DUET-Rel	DUET	Δ	p
<i>Step 15, medium</i> ($N=500$)					
	Spatial	91.2	96.8	+5.6	3×10^{-5}
	Object	97.4	97.0	-0.4	0.84
	Goal	80.8	88.2	+7.4	5×10^{-4}
	LIBERO-10	64.2	86.6	+22.4	1×10^{-17}
	Overall	83.4	92.2	+8.8	—
<i>Step 23, medium+severe</i> ($N=4000$)					
	Overall	73.5	90.8	+17.3	—

the introduction.

5) *Contact and Scene Interaction*: We apply an external force during contact, perturbing robot-world dynamics without resetting the end-effector pose. Across eight suite-severity groups, DUET reaches 95.95% versus 92.22% for DUET-Rel (+3.73 points; Table VI). Every cell is non-negative. The largest lifts are on Goal (+7.6 and +9.6 points), where drawer and plate interactions amplify short contact disturbances. Relative control is already strong on Spatial and Object ($> 96\%$), so the aggregate margin is smaller than under action or teleport failures; the result still shows that dual-action retry does not destabilize contact and

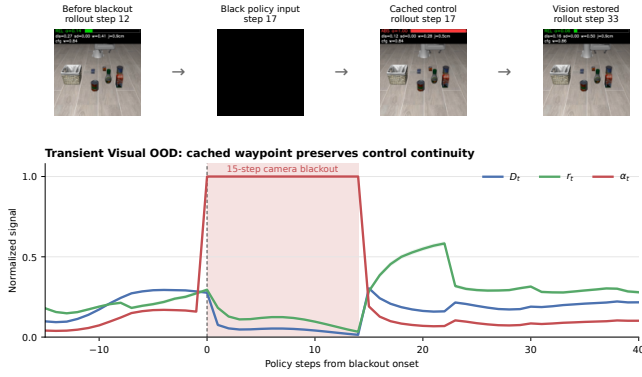


Fig. 5. Camera blackout on the Object suite. DUET holds the last reliable absolute waypoint while the image is invalid and resumes adaptive dual-action control when vision returns. The lower panel aggregates 500 episodes aligned to blackout onset.

TABLE VI

CLOSED-LOOP SUCCESS (%) UNDER EXTERNAL-FORCE CONTACT DISTURBANCES (FIXED-STEP, 100% INJECTION, $N=500$ PER CELL).

Suite	Medium			Severe		
	DUET-Rel	DUET	Δ	DUET-Rel	DUET	Δ
Spatial	96.6	98.4	+1.8	96.8	96.8	0.0
Object	97.6	99.0	+1.4	96.2	96.8	+0.6
Goal	89.2	96.8	+7.6	87.2	96.8	+9.6
LIBERO-10	87.8	92.6	+4.8	86.4	90.4	+4.0
Overall	DUET-Rel 92.22 DUET 95.95 $\Delta = +3.73$					

helps on the more interaction-rich suite.

E. Ablations

The ablations explain which components support (c1)–(c3). Table VII therefore separates inference-time controls on the shared dual-head checkpoint from training-time replacements that require a new run. We report clean success together with the two disturbances where relative control loses a return target—robot-arm teleport and action-execution corruption.

Disabling the absolute branch (DUET-Rel, $\alpha_t = 0$) is the completed inference ablation. It accounts for the closed-loop gains already reported: clean success falls from 96.9% to 93.8%, teleport from 61.7% to 40.0%, and action-execution from 64.6% to 27.9%. Holding the absolute CFG scale fixed at $w=1$ on Object leaves performance essentially unchanged (98.2% clean vs. 99.4%; 83.2% vs. 84.4% under action-medium). The retry benefit therefore comes from using the absolute waypoint, not from adaptive guidance scaling. Absolute-only control ($\alpha_t = 1$) and fixed fusion ($\alpha_t = 0.5$) reuse the same checkpoint and will test whether a constant mix can replace the disagreement gate; those runs are not yet complete.

Training ablations need matched 10k checkpoints. Replacing RVQ with an MLP tests discrete waypoint anchoring; training the relative head on perturbed images tests whether asymmetric augmentation is needed to keep the two streams complementary; and giving the absolute head proprioception

tests the state-free global reference. These rows are preregistered and left as TODO.

F. Real-Robot Evaluation

We reserve a real-robot evaluation on a Franka manipulator covering free-space manipulation, constrained contact, and sustained surface contact. The three tasks are pick-and-place, drawer opening, and whiteboard wiping. Each task will be evaluated under nominal execution and a controlled end-effector displacement that preserves the task objective. The reserved table reports nominal success, disturbed success, injection coverage, and retry success; these experiments test whether (c2) and (c3) transfer beyond simulation.

For each task, we evaluate nominal execution and a controlled end-effector displacement that leaves the scene and task objective unchanged. The displacement is injected before grasp acquisition for pick-and-place, during the final handle approach for drawer opening, and during the contact stroke for whiteboard wiping. This common intervention isolates trajectory re-entry while exposing the controller to free-space, constrained, and sustained-contact regimes. The task-specific displacement direction and magnitude are fixed before evaluation and measured from Franka forward kinematics rather than estimated from video.

Injection occurs only after a predefined visual or contact precondition and must remain within prespecified workspace and force limits. A trial is marked as injected when these criteria are satisfied at intervention time; no trial is excluded based on its subsequent outcome. Trials that never reach the precondition are reported separately as uninjected rather than counted as retry failures. Scene resets use physical markers or fixtures, and DUET and DUET-Rel trials are interleaved in randomized blocks to reduce confounding from hardware drift. Outcomes are scored using predefined task criteria:

- **Pick-and-place:** the commanded object is lifted and deposited fully inside the target receptacle.
- **Drawer opening:** the robot acquires the correct handle and moves the drawer beyond a prespecified displacement threshold.
- **Whiteboard wiping:** the robot maintains safe contact and removes at least a prespecified fraction of a standardized marked region.

In addition to overall task success, we report injection coverage and *retry success*: completion among trials satisfying the prespecified injection criteria. We report successes/trials with Wilson 95% confidence intervals rather than hiding the small physical-trial denominator behind percentages. A hardware safety stop is always active but is not used as DUET’s failure signal; safety-stop events are reported as failures.

G. Discussion

The experiments are organized around three claims. For (c1), cross-action disagreement provides an endogenous anomaly signal without failure examples. For (c2), enabling the absolute branch preserves nominal task success. For (c3), adaptive waypoint fusion improves closed-loop completion after recoverable execution disturbances, especially when

TABLE VII

COMPONENT ABLATIONS, SUCCESS (%). FULL DUET AND DUET-REL ARE FOUR-SUITE AVERAGES (CLEAN $N=2000$; ROBOT-STATE TELEPORT MEDIUM $N=2000$; ACTION-EXECUTION MEDIUM+SEVERE $N=4000$). ENTRIES MARKED $\dagger$ ARE OBJECT-ONLY ($N=500$); THE MATCHED OBJECT FULL-DUET RATES ARE 99.4% CLEAN AND 84.4% ACTION-EXECUTION MEDIUM.

Type	Variant	Checkpoint	Clean	Robot-state	Action-execution	Isolates
Inference	Full DUET	shared	96.9	61.7	64.6	Complete controller
	DUET-Rel ($\alpha = 0$)	shared	93.8	40.0	27.9	Absolute retry branch
	Absolute-only ($\alpha = 1$)	shared	TODO	TODO	TODO	Cost of losing local control
	Fixed fusion ($\alpha = 0.5$)	shared	TODO	TODO	TODO	Adaptive switching
	No adaptive CFG $\dagger$	shared	98.2	TODO	83.2	CFG scale schedule
Training	RVQ $\rightarrow$ MLP	retrained	TODO	TODO	TODO	Discrete waypoint anchoring
	Symmetric augmentation	retrained	TODO	TODO	TODO	Complementary specialization
	State-aware ABS	retrained	TODO	TODO	TODO	State-free global reference

TABLE VIII

REAL-ROBOT EVALUATION. SUCCESS ENTRIES ARE REPORTED AS SUCCESSES/TRIALS (%); RETRY SUCCESS CONDITIONS ONLY ON TRIALS MEETING THE PRESPECIFIED INJECTION CRITERIA. VALUES AND DISTURBANCE DEFINITIONS ARE PLACEHOLDERS PENDING PHYSICAL EXPERIMENTS.

Task	Recoverable disturbance	Coverage	Nominal success		Disturbed success		Retry success		
			DUET-Rel	DUET	DUET-Rel	DUET	DUET-Rel	DUET	Δ
Pick-and-place	EEF shift before grasp (TODO cm)	TODO	TODO	TODO	TODO	TODO	TODO	TODO	TODO
Drawer opening	EEF shift before contact (TODO cm)	TODO	TODO	TODO	TODO	TODO	TODO	TODO	TODO
Whiteboard wiping	EEF shift during stroke (TODO cm)	TODO	TODO	TODO	TODO	TODO	TODO	TODO	TODO
Overall	—	TODO	TODO	TODO	TODO	TODO	TODO	TODO	TODO

relative control loses a return direction. The real-robot evaluation remains reserved for future completion.

V. LIMITATIONS

DUET assumes access to absolute state labels aligned with the relative action chunks. This is natural in LIBERO and many robot logs, but requires accurate calibration and state estimation on the Franka platform. The absolute waypoint is an RVQ demonstration anchor, not an object tracker: a world-state shift can invalidate the grasp reference even when the robot remains nearby. The bound-violation term S_t is uninformative on LIBERO, so robot-state gating relies on the residual to the predicted waypoint; this still cannot distinguish “the arm left a valid target” from “the target itself moved.” Complete visual dropout requires an explicit validity guard and a cached waypoint rather than disagreement-based detection. Matched component ablations, detection analysis, and Franka results remain incomplete in this draft.

VI. CONCLUSION

We presented DUET, a dual-branch VLA policy for endogenous failure detection and trajectory re-entry. By aligning local relative actions with stateless, RVQ-anchored absolute waypoints, DUET uses cross-space disagreement to determine when execution has left learned behavior and uses the waypoint correction to retry it. The resulting controller preserves nominal performance and substantially improves completion under recoverable pose and action-execution deviations without failure labels or takeover demonstrations. Its scope is deliberately narrower than general recovery planning: stale world references, irreversible failures, and

fully invalid observations require additional mechanisms or a new task strategy.

ACKNOWLEDGMENT

TODO: Add acknowledgments, funding, compute resources, and any required dataset or codebase acknowledgments after the author list is finalized.

REFERENCES

- [1] A. Brohan, N. Brown, J. Carbajal, Y. Chebotar, X. Chen, K. Choro-manski, T. Ding, D. Driess, A. Dubey, C. Finn, P. Florence, C. Fu, M. G. Arenas, K. Gopalakrishnan, K. Han, K. Hausman, A. Herzog, J. Hsu, B. Ichter, A. Irpan, N. Joshi, R. Julian, D. Kalashnikov, Y. Kuang, I. Leal, K.-H. Lee, S. Levine, Y. Lu, H. Michalewski, I. Mordatch, K. Pertsch, K. Rao, K. Reymann, M. Ryoo, G. Salazar, P. Sanketi, P. Sermanet, J. Singh, A. Singh, R. Soricut, H. Tran, V. Vanhoucke, Q. Vuong, A. Wahid, S. Welker, P. Wohlhart, J. Wu, F. Xia, T. Xiao, P. Xu, S. Xu, T. Yu, and B. Zitkovich, “RT-2: Vision-language-action models transfer web knowledge to robotic control,” *arXiv preprint arXiv:2307.15818*, 2023.
- [2] M. J. Kim, K. Pertsch, S. Karamcheti, T. Xiao, A. Balakrishna, S. Nair, R. Rafailov, E. Foster, T. Lampe, P. Sanketi, Q. Vuong, T. Kollar, B. Burchfiel, R. Tedrake, D. Sadigh, C. Finn, and P. Liang, “OpenVLA: An open-source vision-language-action model,” *arXiv preprint arXiv:2406.09246*, 2024.
- [3] StarVLA Community, “StarVLA: A lego-like codebase for vision-language-action model developing,” <https://github.com/starVLA/starVLA>, 2026.
- [4] C. Xu, T. K. Nguyen, E. Dixon, C. Rodriguez, P. Miller, R. Lee, P. Shah, R. Ambrus, H. Nishimura, and M. Itkina, “Can we detect failures without failure data? uncertainty-aware runtime failure detection for imitation learning policies,” in *Proceedings of Robotics: Science and Systems*, 2025.
- [5] S. Ross, G. Gordon, and D. Bagnell, “A reduction of imitation learning and structured prediction to no-regret online learning,” in *Proceedings of the Fourteenth International Conference on Artificial Intelligence and Statistics*, 2011.

- [6] H. Liu, S. Nasiriany, L. Zhang, Z. Bao, and Y. Zhu, “Robot learning on the job: Human-in-the-loop autonomy and learning during deployment,” in *Proceedings of Robotics: Science and Systems*, 2023.
- [7] R. Hoque, A. Mandlekar, C. R. Garrett, K. Goldberg, and D. Fox, “IntervenGen: Interventional data generation for robust and data-efficient robot imitation learning,” in *IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2024, pp. 2840–2846.
- [8] S.-W. Lee, X. Kang, and Y.-L. Kuo, “Diff-DAGger: Uncertainty estimation with diffusion policy for robotic manipulation,” in *IEEE International Conference on Robotics and Automation*, 2025.
- [9] C. Agia, R. Sinha, J. Yang, Z. Cao, R. Antonova, M. Pavone, and J. Bohg, “Unpacking failure modes of generative policies: Runtime monitoring of consistency and progress,” in *Proceedings of the Conference on Robot Learning*, ser. Proceedings of Machine Learning Research, vol. 270, 2025, pp. 689–723.
- [10] R. Römer, A. Kobras, L. Worbis, and A. P. Schoellig, “Failure prediction at runtime for generative robot policies,” in *Advances in Neural Information Processing Systems*, 2025.
- [11] Q. Gu, Y. Ju, S. Sun, I. Gilitschenski, H. Nishimura, M. Itkina, and F. Shkurti, “SAFE: Multitask failure detection for vision-language-action models,” in *Advances in Neural Information Processing Systems*, vol. 38, 2025.
- [12] J. Duan, W. Pumacay, N. Kumar, Y. R. Wang, S. Tian, W. Yuan, R. Krishna, D. Fox, A. Mandlekar, and Y. Guo, “AHA: A vision-language-model for detecting and reasoning over failures in robotic manipulation,” in *International Conference on Learning Representations*, 2025.
- [13] J. Spencer, S. Choudhury, M. Barnes, M. Schmittle, M. Chiang, P. Ramadge, and S. Srinivasa, “Learning from interventions: Human-robot interaction as both explicit and implicit feedback,” in *Proceedings of Robotics: Science and Systems*, 2020.
- [14] Y. Dai, J. Lee, N. Fazeli, and J. Chai, “RACER: Rich language-guided failure recovery policies for imitation learning,” in *IEEE International Conference on Robotics and Automation*, 2025, pp. 15 657–15 664.
- [15] H. Chen, Y. Yao, R. Liu, C. Liu, and J. Ichnowski, “Automating robot failure recovery using vision-language models with optimized prompts,” *arXiv preprint arXiv:2409.03966*, 2024.
- [16] C. Chi, S. Feng, Y. Du, Z. Xu, E. Cousineau, B. Burchfiel, and S. Song, “Diffusion policy: Visuomotor policy learning via action diffusion,” *The International Journal of Robotics Research*, 2024.
- [17] K. Black, N. Brown, D. Driess, A. Esmail, M. Equi, C. Finn, N. Fusai, L. Groom, K. Hausman, B. Ichter, J. Jakubczak, T. Jones, A. Khazatsky, S. Levine, A. Li-Bell, Y. Lu, S. Nair, K. Pertsch, Q. V. Xu, F. Xia, and T. Xiao, “ π_0 : A vision-language-action flow model for general robot control,” *arXiv preprint arXiv:2410.24164*, 2024.
- [18] A. van den Oord, O. Vinyals, and K. Kavukcuoglu, “Neural discrete representation learning,” in *Advances in Neural Information Processing Systems*, 2017.
- [19] N. Zeghidour, A. Luebs, A. Omran, J. Skoglund, and M. Tagliasacchi, “Soundstream: An end-to-end neural audio codec,” *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, 2022.
- [20] J. Ho and T. Salimans, “Classifier-free diffusion guidance,” *arXiv preprint arXiv:2207.12598*, 2022.
- [21] Y. Burda, H. Edwards, A. Storkey, and O. Klimov, “Exploration by random network distillation,” in *International Conference on Learning Representations*, 2019.